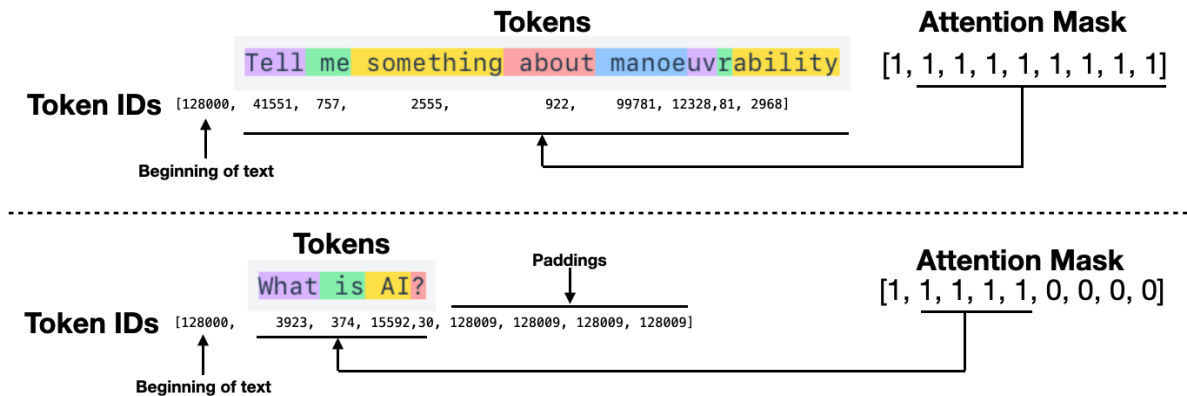
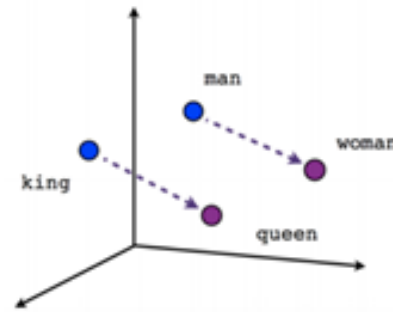


Tokenization: From Text to Numerical IDs

- Tokenization is the process of breaking down raw text into smaller units called "tokens" so computers can process them.
- **Word Tokenization:** Each unique word is assigned a specific integer ID.
- **Character Tokenization:** Each individual letter is treated as a token, which helps with spelling but loses semantic meaning.
- **BPE (Byte Pair Encoding):** A sub-word tokenization method used by modern LLMs to handle rare words and prevent "out-of-vocabulary" errors.



Word Vectors: Turning Words into Mathematics



Male-Female



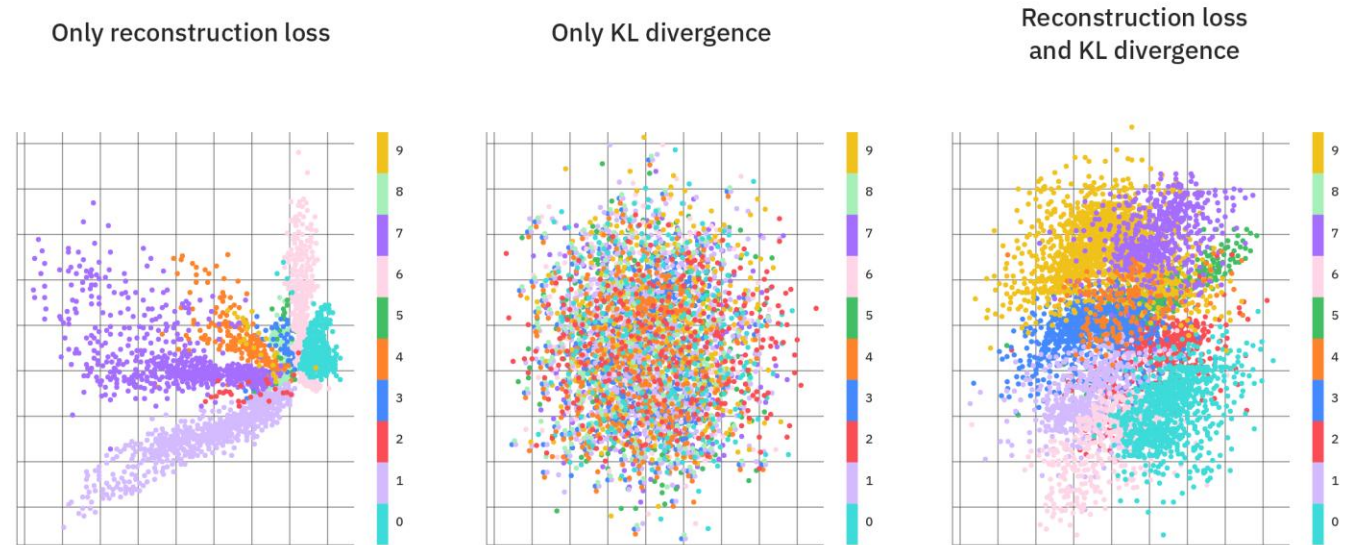
Verb tense



Country-Capital

- Word Vectors represent text as coordinates in high-dimensional space.
- **Word2Vec & GloVe**: These are traditional algorithms that turn words into math based on their neighboring words in a dataset.
- Unlike simple IDs, these vectors capture semantic relationships (e.g., the distance between "King" and "Queen" is similar to "Man" and "Woman").

Latent Space: Visualizing Meaning



- The "Latent Space" is the mathematical map where all word vectors live.
- AI understands context by placing related words like "Doctor" and "Hospital" close to each other mathematically.
- By calculating the distance between vectors, the AI can determine how similar two pieces of text are.